Comparative Efficacy of Antidepressants for Symptoms Remission of Gastroesophageal Reflux: A Bayesian Network Meta-Analysis of Randomized Controlled Trails (protocol)

Xiao-Bei Si[1], De-Ying Bi[1], Lin-Yu Huo[2], Yu Lan[1*], Shuo Zhang [3]

[1] Department of Gastroenterology, Beijing Jishuitan Hospital, Beijing 100035, China
[2] Department of Neurology, Beijing Haidian Hospital, Beijing 100080, China
[3] Department of Cardiology, Fu Wai Hospital, National Center for Cardiovascular Diseases, Chinese Academy of Medical Sciences, Beijing 100038, China
*Corresponding author

Abstract

Objective The aim of this study is to compare symptoms' remission of antidepressants for gastroesophageal reflux.

Methods We plan to include randomized controlled trials (RCTs) that evaluate efficacy of antidepressants for gastroesophageal reflux by searching the MEDLINE, EMBASE, Web of Science, China National Knowledge Infrastructure Database (CNKI), Chinese VIP Information (VIP), Chinese Medical Databases (CMB) and Wan-Fang databases from the starting date of database establishment till September 31, 2019. The words of "gastroesophageal reflux disease", "antidepressant" and their synonyms will be adopted as searching terms. ADDIS 1.16.8 will be used for network meta-analysis, with conduction of node split analysis as well as rank probability analysis.

Conclusions The present analysis aims to further compare efficacy of symptom relieving of gastroesophageal reflux among antidepressants, which apply clinical evidence for further therapy of gastroesophageal reflux diseases.

Key words: antidepressants; gastroesophageal reflux; network meta-analysis

Gastro-esophageal reflux symptoms stand for a group of clinical syndromes including acid-reflux, heartburn, chest pain, throat discomfort, cough, and epigastric discomfort [1,2]. In recent years, gastro-oesophageal reflux symptoms was no longer confined to gastroesophageal reflux disease (GERD), but is thought to originate from a variety of diseases, including GERD, functional heartburn (FH) and reflux hypersensitivity (RH) [2]. Previous epidemiological studies have shown that the incidence of gastro-oesophageal reflux symptoms in the Chinese population was 12.6% [3] with increasing trend in recent years [3].

Psychosocial disorders such as anxiety and depression may be the co-causative factors of the above-mentioned gastroesophageal reflux symptoms [4], and the use of antidepressants may be effective to remission of gastroesophageal reflux symptoms [5]. In recent years, a series of randomized controlled trials have been conducted with antidepressants therapy aiming to improve gastroesophageal reflux symptoms [6]. Kinds of antidepressants, such as selective serotonin uptake inhibitors (SSRIs) [7], tricyclic antidepressants (TCAs) [8], etc., were referred with conclusions these antidepressants might have a role in improving gastroesophageal reflux symptoms. Accordingly, we plan to conduct such network meta-analysis aiming to further compare the efficacy of antidepressants in remission of esophageal reflux symptoms based on previous randomized controlled trials.

MATERIALS AND METHODS

Search Strategy

We plan to search the databases of MEDLINE, EMBASE, the Web of Science, the Cochrane Database, the China National Knowledge Infrastructure Database, Chinese VIP Information Databases, Chinese Medical Databases, and the Wan-Fang Database using the search terms of "Gastroesophageal reflux disease", "antidepressant", "randomized controlled trials" and their synonyms. The search was limited to the studies up to September 31, 2019. The search strategies are provided in Column 1 with Medline as an example.

Inclusion and Exclusion Criteria

The inclusion criteria of present study are as follows: (1) The participants enrolled in the study should be of GERD and/or FH and/or RH, with or without psychological disorders such as anxiety and depression. (2) Participants in experimental group were treated with antidepressants and (3) the participants in control group were treated with placebo or other types of antidepressants. (4) The outcome should be total remission rate of gastroesophageal reflux symptoms, i.e., the total remission rate = number of patients with remission of at least one symptom/total number × 100%. (5) The study type should be randomized controlled trial.

The exclusion criteria are as follows: (1) duplicate publications; (2) literature reviews; (3) non-randomized trials or trials with mistake randomization method also stated to be "randomized"; (4) Case reports.

Data Extraction and Quality Evaluation

All retrieved studies will be independently screened by two reviewers. Titles and abstracts will be screened for all relevant articles. Full texts will be screened for further assessments according to the inclusion and exclusion criteria. Disagreements will be resolved by discussion or through consultation with a second specialist.

Studies that met the inclusion criteria will be independently reviewed by two reviewers independently. Data extracted from the included trials include authors, publication year, sample size, interventions and outcome.

Quality evaluation

The methodological quality of each trial will be evaluated based on the Cochrane handbook. Methodological quality will be assessed with regard to random sequence generation, allocation concealment, the blinding of participants and personnel, incomplete outcome data, the blinding of outcome assessments, selective reporting, and other bias. Each item will be assessed based on the following responses: “high risk of bias”, “low risk of bias” and “Unclear”.

Statistical analysis

ADDIS software (version 1.16.5) will be adopted to perform all the statistical analyses based on Bayesian framework. The priori evaluation and data processing are conducted using the Markov chain Monte Carlo algorithms. The consistency test will be performed using the inconsistency standard deviation (ISD) and node-splitting analysis. For the closed-loop index, both analyses are used. On the contrary, only ISD will be used for the open-loop indicators. For ISD, if the range of 95% CI of ISD includes, the consistency model will be used, otherwise the inconsistency model will be used. For node-splitting analysis, If the P value of the node-splitting analysis more than 0.05, the consistency model will be used to calculate the pooled effect size. Otherwise, the nonconsistency model will be used. The convergence of the model will be determined by the potential scale reduction factor (PSRF) of the Brooks–Gelman–Rubin method.

PSRF that closed to 1 indicate good convergence and may be accepted. The Parameters for the ADDIS software set as follows: number of chains, 4; tuning iterations, 20,000; simulation iterations, 50,000; thinning interval, 10; inference samples, 10,000; and variance scaling factor, 2.5.[9]

DISCUSSION

Previous studies have shown that the medication of antidepressants might be beneficial for gastro-esophageal reflux symptoms. In 2015, Weijenborg et al. [10] performed a meta-analysis to compare the effects of antidepressants for esophageal dysfunction/gastro-esophageal reflux disease with placebo as control. As a result, antidepressants showed higher remission rate than placebo especially for symptoms of heartburns. What's more, SSRIs might be superior to TCAs according to subgroup analysis. In the same year, another meta-analysis from Lin et al. [11] compared the remission of GERD symptoms between routine treatment plus SSRIs (or SSRI alone) and GERD routine treatment. As a result, SSRIs showed higher remission rate of GERD symptoms than that of control. In the following year, a meta-analysis by Zou et al. [12] showed that FM combined with acid suppression treatment significantly improved gastroesophageal reflux symptoms compared with acid suppression alone. However, this study included only randomized placebo-controlled trials but not GERD routine treatment-controlled trials. As a result, Placebo comparison showed that TCAs and SSRIs showed significantly higher remission rate of gastro-esophageal reflux symptoms than placebo. Such results were in consistence with previous studies.

The present meta-analysis will be the first network meta-analysis comparing antidepressants for symptom relieving of gastroesophageal reflux. There may be some limitations to the present study. First, as we aimed to explore the antidepressants in patients with gastro-esophageal reflux symptoms. Thus, the studies with subjects of GERD, FH and RH will be included. Such difference contributes to bias for further meta-analysis. Second, we plan to include placebo-controlled and antidepressants-controlled studies. Such criteria may limit the number of included studies.

CONCLUSIONS

The present analysis aims to further compare efficacy of symptom relieving of gastroesophageal reflux among antidepressants, which apply clinical evidence for further therapy of gastroesophageal reflux diseases.

Conflict of interest statement

The authors declare no conflict of interest.

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Column 1 Search strategy

#1	Gastroesophageal reflux disease[title/abstract] OR Gastroesophageal Reflux[title/abstract] OR Gastroesophageal Reflux[MeSH] OR non-erosive reflux disease[title/abstract] OR GERD[title/abstract] OR NERD[title/abstract] OR reflux esophagitis[title/abstract] OR RE[title/abstract] OR Barrett Esophagus[title/abstract] OR Barrett Esophagus[MeSH] OR heartburn[title/abstract] OR heartburn[MeSH] OR esophageal hypersensitivity[title/abstract] OR reflux hypersensitivity[title/abstract]
#2	5-HT reabsorption inhibitor[title/abstract] OR 5-HT and norepinephrine reabsorption inhibitor[title/abstract] OR monoamine oxidase inhibitor[title/abstract] OR Monoamine Oxidase Inhibitors[MeSH] OR tetracyclic[title/abstract] OR benzodiazepines[title/abstract] OR benzodiazepines[MeSH] OR SSRI[title/abstract] OR TCA[title/abstract] OR BZD[title/abstract] OR Delixit[title/abstract] OR tetracyclic[title/abstract] OR OR fitrexin[title/abstract] OR doxepin[title/abstract] OR nortriptyline[title/abstract] OR citalopram[title/abstract] OR fluoxetine[title/abstract] OR imipramine[title/abstract] OR tandospirone[title/abstract] OR diazepam[title/abstract] OR lorazepam[title/abstract]
#3	Randomized controlled trials[title/abstract] OR randomized[title/abstract] OR Randomized Controlled Trials as Topic[MeSH] OR Randomized Controlled Trial [Publication Type]
#4	#1 AND #2 AND #3